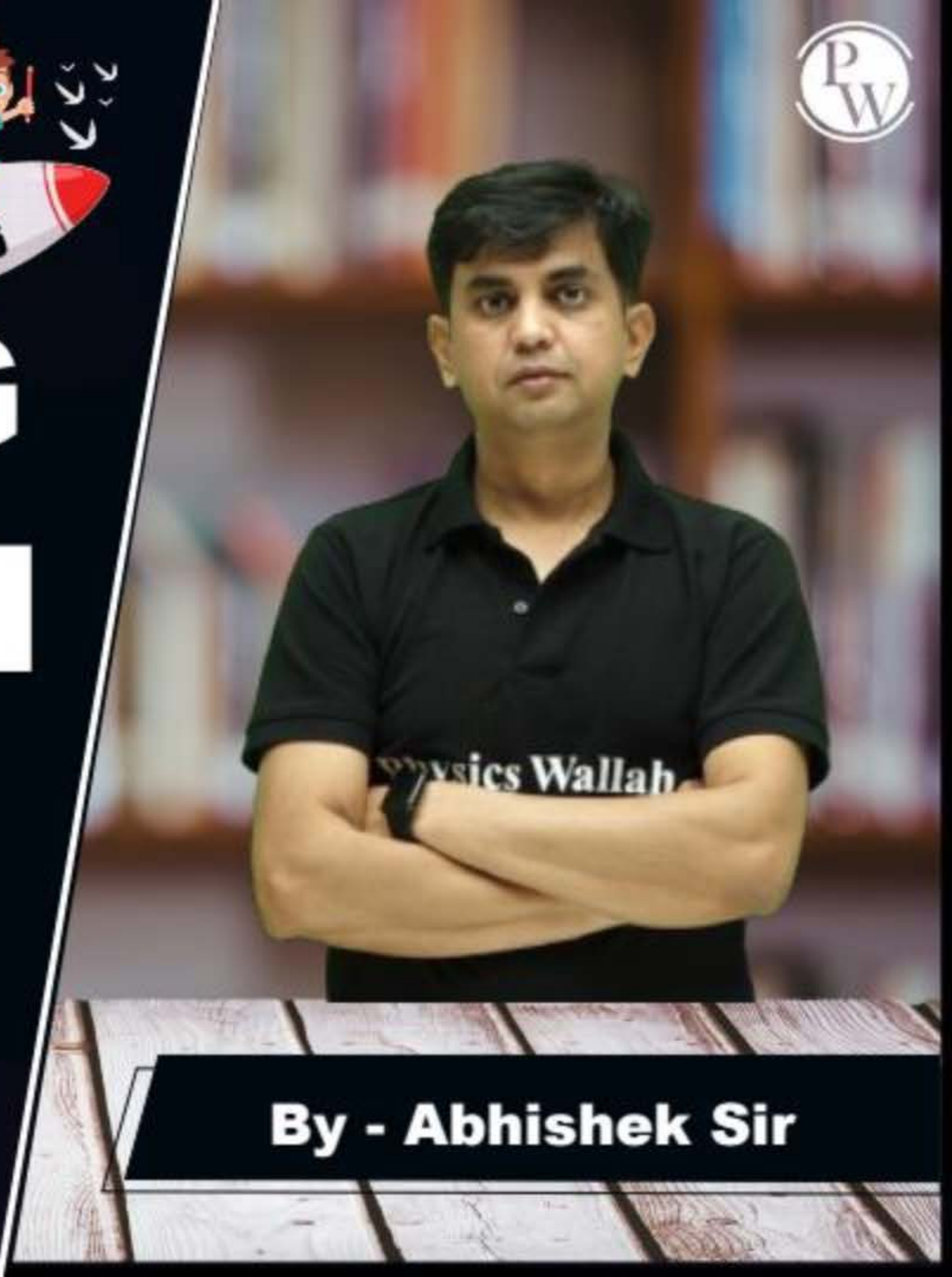
CS & IT ENGINEERING

Computer Network - 1

Transport Layer

Lecture No. – 02



By - Abhishek Sir

Recap of Previous Lecture

Topic

Transport Layer Services

Topics to be Covered

Topic UDP

Topic TCP

Topic : UDP

-> UDP : User Datagram Protocol ⟹ Basic Protocol

-> Provide 'Connection-less' and 'Unreliable' services
[Unordered delivery of messages, messages may be lost]

UDP ⟹ connection less and unreliable

↓

Messages can be delivered in any order


P1
Message 4
Message 3
Message 2
Message 1
UDP
Segment 4
Segment 3
Segment 2
Segment 1
IP
Datagram 4
Datagram 3
Datagram 2
Datagram 1
P2
M3
M4
M1
UDP
S3
S4
S1
IP
A3
A4
A1

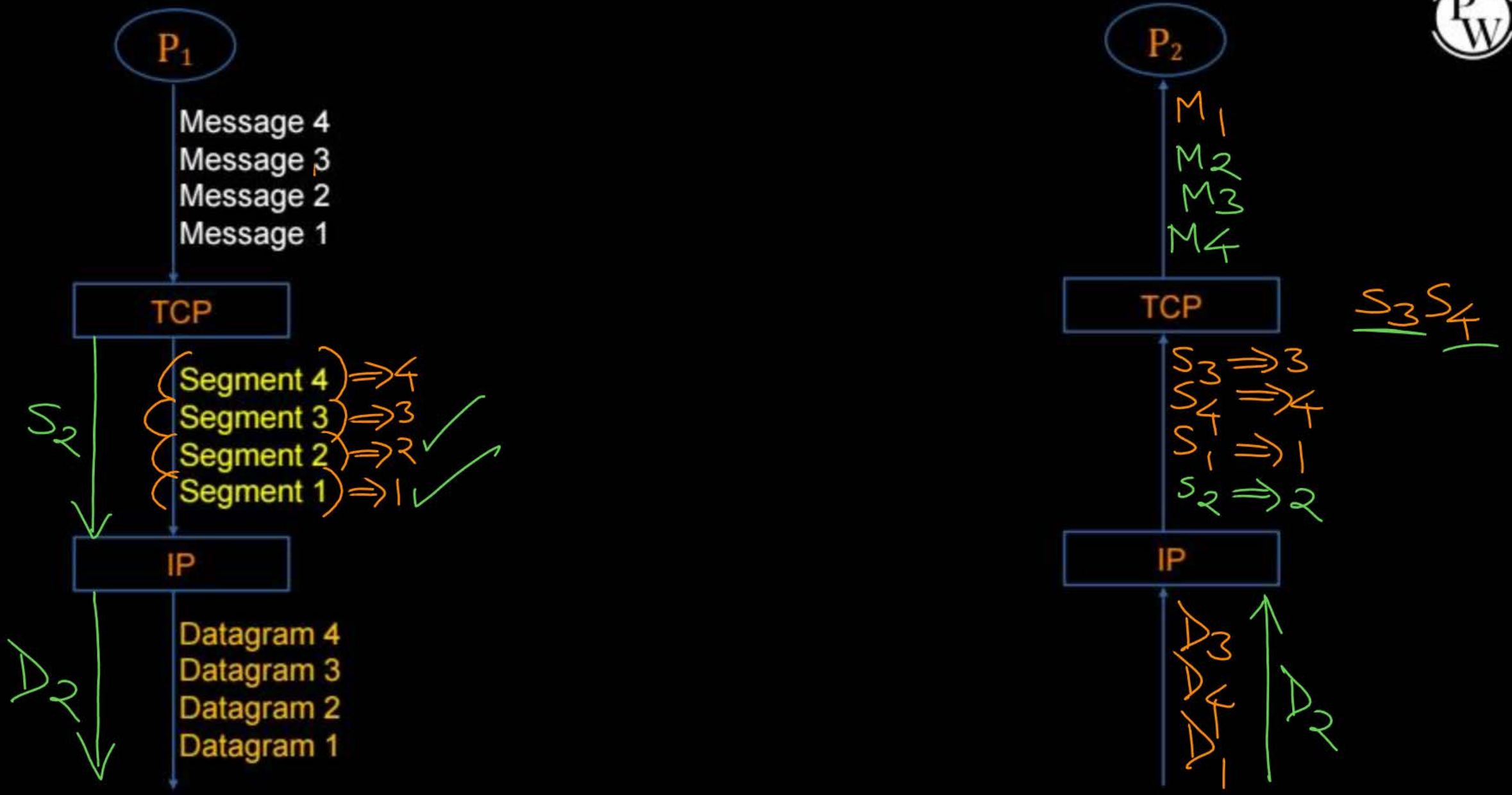
Topic : TCP

-> TCP : Transmission Control Protocol

-> Provide 'Connection-oriented' and 'Reliable' services on top of connection-less and unreliable protocol.
[In-order delivery of messages]

[TCP = UDP + Extra Services]

TCP : Recover the loss, via Retransmission


P1
Message 4
Message 3
Message 2
Message 1
TCP
Segment 4
Segment 3
Segment 2
Segment 1
IP
Datagram 4
Datagram 3
Datagram 2
Datagram 1
P2
TCP
IP

Topic : UDP

=> Connection-less :

-> No handshaking between UDP sender and receiver

-> Each UDP segment handled independently of others

-> Delivery of messages can be any order to the communicating process

=> Unreliable :

-> Messages may be lost

Topic : UDP

=> Simple and Fast :

-> No connection state at sender and receiver

-> Small (8 bytes, fixed) header size

-> No any flow and congestion control

Topic : UDP

-> UDP : User Datagram Protocol

-> Prefer for shorter communication [like query and response]

-> UDP as transport protocol used by :

DNS, SNMP, HTTP/3, RIP and Real Time Multimedia [Streaming Multimedia]

Topic : Demultiplexing

Demultiplexing at receiver :

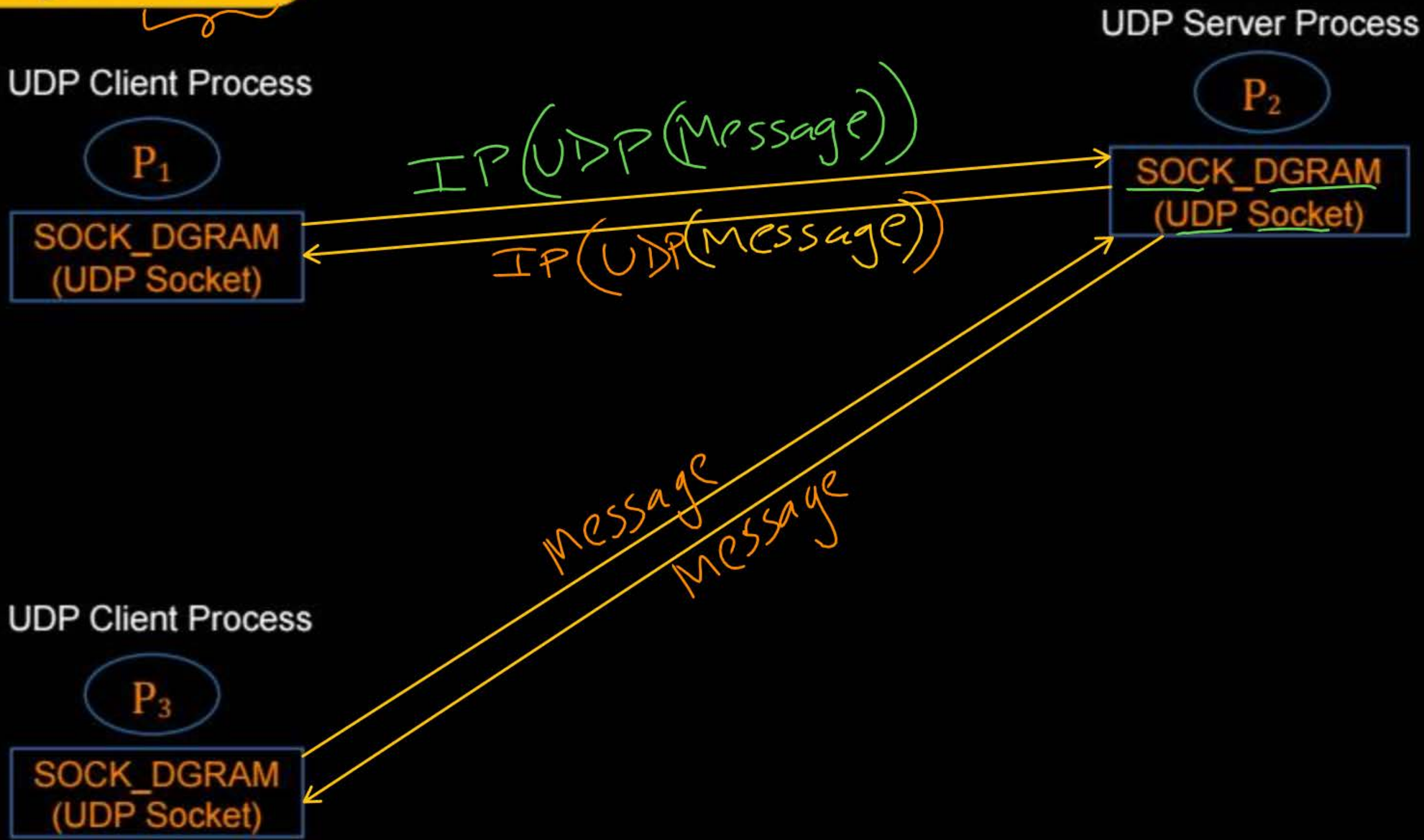
1. Connection-less demultiplexing
2. Connection-oriented demultiplexing

Topic : Connection-Less Demultiplexing

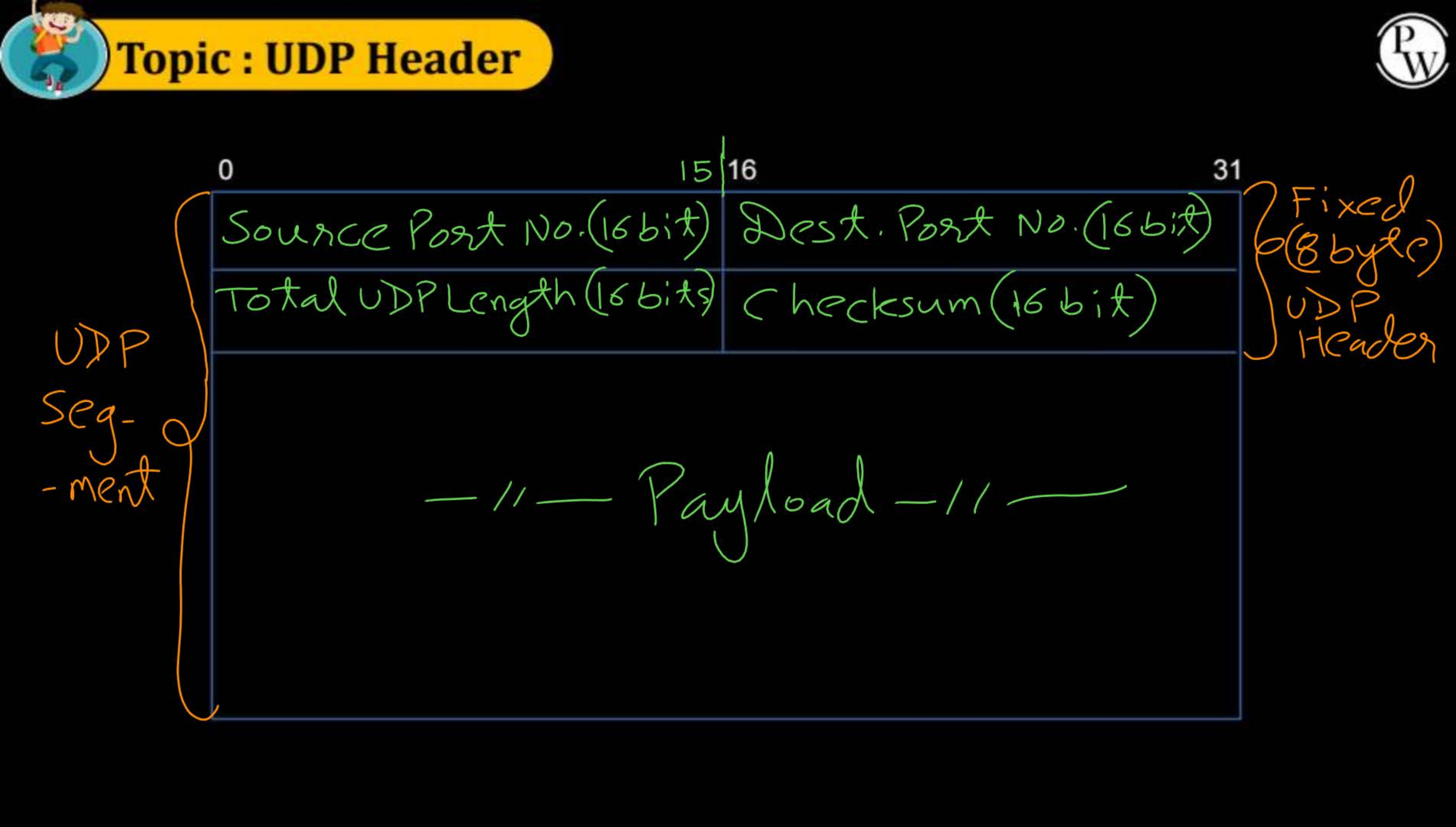
-> UDP sockets identified by 2 tuples :

Destination IP address and Destination Port Number

Topic : UDP


UDP Server Process
UDP Client Process
P1
P2
SOCK_DGRAM
(UDP Socket)
IP(UDP(Message))
IP(UDP(Message))
SOCK_DGRAM
(UDP Socket)
Message
Message
UDP Client Process
P3
SOCK_DGRAM
(UDP Socket)

Topic : UDP Header



Topic : UDP Header

=> Source port number : 16-bits

=> Destination port number : 16-bits

=> Total Length : 16-bits

-> Size of UDP segment in bytes

-> Including 8 bytes header size

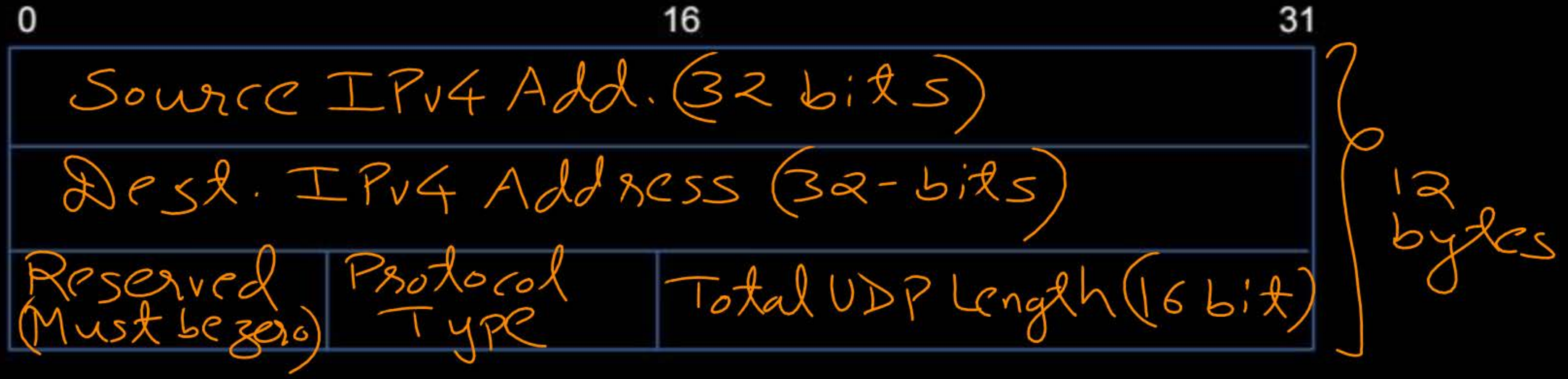
-> Maximum UDP segment size = $(2^{16} - 1)$ bytes

-> Maximum UDP payload size = $(2^{16} - 9)$ bytes

UDP Payload size = (Total length − 8) bytes

Topic : IPv4 Pseudo-Header

$\Rightarrow$ size = 12 bytes



Topic : IPv4 Pseudo-Header

=> Source IP Address : 32-bits

=> Destination IP Address : 32-bits

=> Reserved : 8-bits

-> Must be zero

=> Protocol : 8-bits

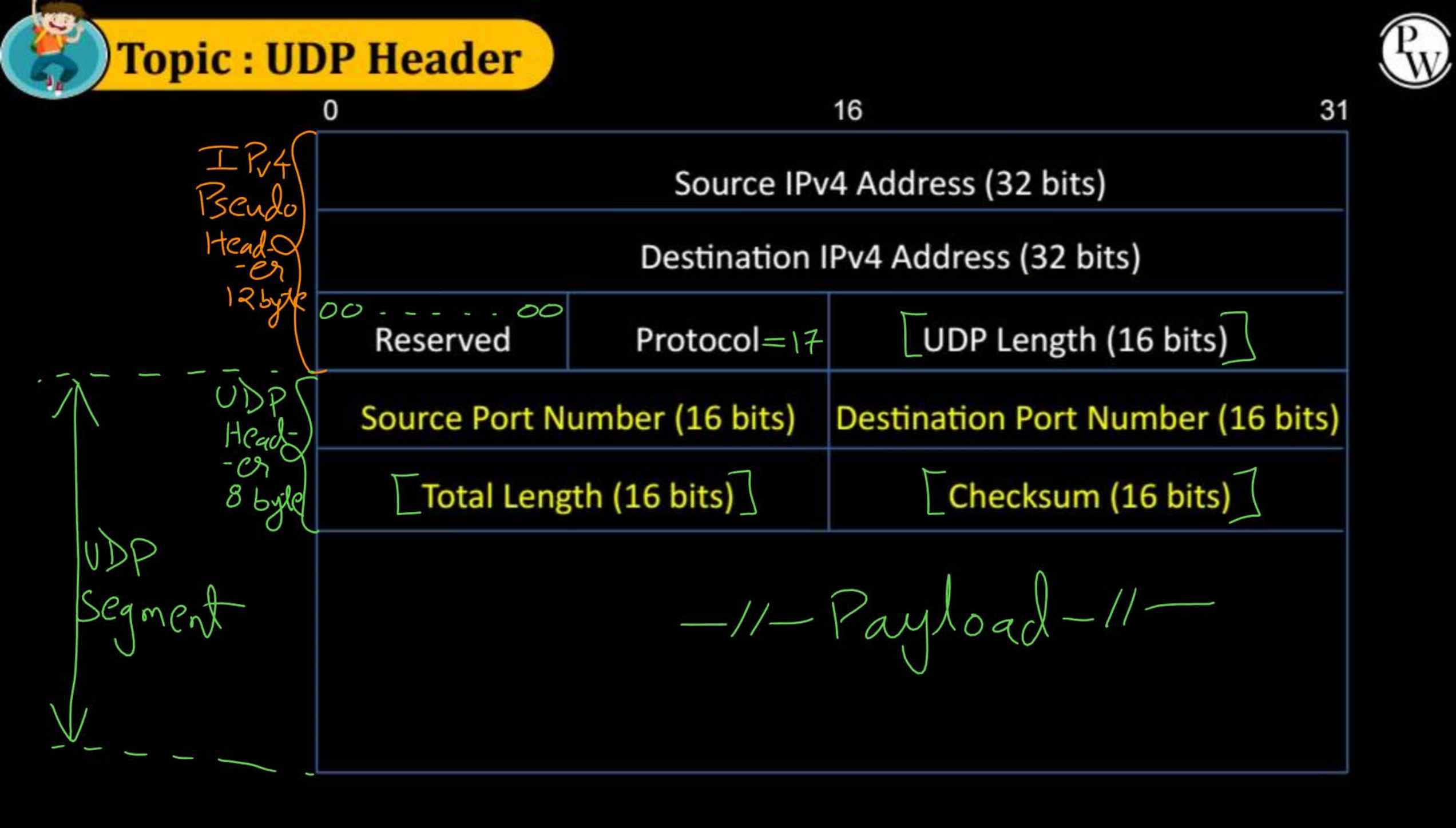
-> Same as IP protocol type

=> UDP Length : 16-bits

-> Size of UDP segment in bytes

-> 8 bytes header size plus payload size

Topic : UDP Header



Topic : UDP Header

=> Checksum : 16-bits

-> 16-bit one's complement of the one's complement sum of all 16-bit words in UDP segment including pseudo header

-> Error detection is optional in UDP, it depends upon the application

-> If application wants to disable error detection process then sender set all the 16 bits are zero of checksum field

Topic : TCP

=> Connection-oriented :

-> Need to establish (logical) connection between TCP sender and receiver

- 3-way handshaking for connection establishment
- 4-way handshaking for connection close

-> In order delivery of messages to the communicating process

Topic : TCP

=> Reliable :

-> Recover lost messages, via retransmission

=> Complex and Slow :

-> Need to maintain connection state at TCP sender and receiver

-> Provide flow and congestion control

Topic : TCP

-> TCP : Transmission Control Protocol

-> Full-duplex and point-to-point logical connection

-> Provide Flow, Error and Congestion Control

-> Prefer for long communication

2 mins Summary

Topic UDP

Topic TCP

THANK - YOU